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Determination of phenyl urea pesticides in apple juice by HPLC-DAD employing sol-gel carbowax 20 M modified fabric phase sorptive extraction membranes

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ABSTRACT

Fabric phase sorptive extraction (FPSE) has established itself as a powerful tool for pesticide residue monitoring in food analysis. In this work, FPSE employing sol-gel carbowax 20 M modified cellulose membranes was combined with high performance liquid chromatography-diode array detection (HPLC-DAD) for the determination of a number of widely used phenyl urea herbicides (i.e., chlorbromuron, isoproturon, monolinuron, and linuron) in fruit juice samples for the first time. The choice of the extraction medium, extraction time, stirring rate, sample volume, and ionic strength, as well as the elution solvent type and volume and elution time were optimized. The proposed method was validated in terms of linearity, trueness, sensitivity, selectivity, and precision. The limit of detection was found to be 0.6 ng mL^{-1} for all investigated pesticides. Moreover, the trueness, expressed as recovery rate, ranged from 84.5 to 109.8 % for intra-day and from 94.5 to 113.5 % for intra-day measurements. The proposed method exhibited a green character and good practicality, while the FPSE membranes were found to be reusable for at least 20 times. Finally, the developed approach was used for the analysis of commercial samples, demonstrating the feasibility of this approach.

1. Introduction

Fruit juices are beverages with a high nutritional value that contain a variety of different vitamins, minerals, organic acids, antioxidants, and fibers. Lately, the beverage market has experienced a notable surge in fruit juice popularity, propelled by an increasingly health-aware consumer base [1]. However, there are many consumers' concerns related to food contaminants, among which pesticide residues in food are the most prominent and therefore play an important factor when purchasing food products [2]. Undoubtedly, the dramatic increase of the world's population in the 20th century and the parallel increase for the requirement of certain food products, significantly enhanced the use of pesticides in agriculture to control weeds and insects [3].

Pesticides can contaminate both surface water and groundwater through different pathways and end up in soil and agricultural crops. Due to the negative health and environmental impacts, the public and policymakers have raised a huge concern about pesticides use in agriculture and their residues in food products [4]. Phenyl urea pesticides (PUHs) are a commonly used class of pesticides that consist of a urea backbone substituted with a phenyl group. PUHs are employed to control herbaceous and perennial weeds, aiding in crop protection. High concentration levels of PUHs can impose toxic effects on birds, aquatic organisms, and humans [5]. Since PUHs can contaminate crops, they may easily be transferred into juices produced from these and pose a risk for the health of consumers [6].

Among the commonly used sample preparation methods for the extraction of pesticides from fruit juice samples are solid-phase extraction (SPE) and liquid-liquid extraction. However, both techniques show big disadvantages due to the large quantities of chemicals that are utilized, the need for multiple operation steps, the preconcentration of the extract required prior to analysis and the potential co-extraction of the interfering compounds [6]. To overcome these significant drawbacks, various microextraction and miniaturized extraction techniques have arisen. The need to develop such approaches is clearly reflected by the principles of Green Analytical Chemistry (GAC) [7] and the principles of Green Sample Preparation (GSP) [8]. GAC concerns the role of analytical

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chemists in making the laboratory practices greener, while GSP is focused on the minimization of the environmental impact of sample preparation. Examples of more environmentally friendly techniques that have been proposed for fruit juice analysis are solid-phase microextraction (SPME) [9], magnetic dispersive solid phase extraction (MDSPE) [10], fabric phase sorptive extraction (FPSE) [11], stir bar sorptive extraction (SBSE) [12], dispersive liquid–liquid microextraction (DLLME) [13], capsule phase microextraction (CPME) [14] etc.

Among these, FPSE is an attractive technique that can be efficiently employed for the sample preparation of foodstuffs [11,15,16]. The microextraction device consists of a natural or synthetic fabric substrate chemically coated with a sol-gel organic-inorganic hybrid sorbent [16]. An important benefit of FPSE is the wide range of available sorbents that can be used to extract numerous analytes of different polarities [15]. Moreover, FPSE is characterized by operational simplicity as well as simplicity in device fabrication, reduced fabrication cost, high enrichment factors, rapid equilibration, ability to perform simultaneous sample preparation of multiple samples, and reusability [17]. Due to its unique properties, this technique has already established itself as a powerful approach for the sample preparation of fruit juices without requiring additional sample pre-treatment steps [11]. Moreover, it has been proven to be a powerful tool for the monitoring of organic pollutants such as organophosphorus pesticides [11], azinphos-methyl and chlorfenvinphos pesticides [18], primicarb and fenitrothion [19], and organochlorine pesticides [20].

The aim of this work was to develop a simple and efficient FPSE protocol combined with HPLC-DAD to monitor PUHs in apple juice samples. For this purpose, sol-gel carbowax 20 M modified cellulose membranes were employed. The extraction and elution steps were investigated and optimized to achieve high extraction efficiency towards the investigated herbicides. Accordingly, the analytical method was validated in terms of sensitivity, linearity, trueness, and precision. The practicality and the greenness of this approach were also studied using Blue Applicability Grade Index (BAGI) [21], Complex Green Analytical Procedure Index (ComplexGAPI) [22], complex modified GAPI (ComplexMoGAPI) [23], and Click Analytical Chemistry Index (CACI) [24]. Finally, the proposed method was used for the analysis of different types of apple juice samples.

2. Experimental

2.1. Instrumentation

For the separation and determination of PUHs, a binary low-pressure gradient Shimadzu HPLC–DAD system (Kyoto, Japan) was used. The system consisted of two LC20-AD pumps, a DGU-20A_3 degassing unit, a SIL-20A autosampler, a CTO-20AC column oven, and a CBM-20A communication bus module. An SPD-M20A DAD was used as a detection system, and data acquisition was performed using the Lab Solutions-LC solutions software.

2.2. Chemicals, reagents, and samples

Acetonitrile (ACN) of HPLC grade, acetone (ACE), ethanol (EtOH), and HPLC grade water were purchased from Honeywell (Charlotte, North Carolina, USA). Methanol (MeOH) of HPLC grade was purchased from Merck (Germany). Chlorbromuron (99.4 %), isoproturon (99.5 %), monolinuron (99.6 %), and linuron (99.6 %) were purchased from Supelco (Bellefonte, PA, USA).

Stock standard solutions of the target analytes were prepared in methanol at a concentration level of 1000 mg L^{-1}. The stock solutions were stored at 4 °C. Working standard solutions were prepared daily through serial dilutions in water.

Six commercially available apple juice samples (two samples of organic and one carbonated juice) were purchased from the local market of Vienna, Austria. No sample pre-treatment step was employed prior to the FPSE protocol. Storage of samples was performed at 4 °C.

The FPSE media consisted of 4 cm^2 (2.0 cm × 2.0 cm) sol–gel Carbowax® 20 M membranes. The preparation of the membranes has been previously described [25].

2.3. Chromatographic conditions

The separation of the PUHs was performed using a Kinetex C_{18} 100 Å (2.6 μm, 4.6 mm × 30 mm) column (Phenomenex, California, USA). The column was thermostated at 30 °C and the injection volume was 5 μL. The mobile phase consisted of H_2O (A) and ACN (B) and the flow rate was set at 1.0 mL min^{-1}. The elution of the pesticides was performed using a gradient starting from 75:25 (A:B, *v*/v) which was kept constant for 2.5 min. Then, the composition of the mobile phase changed to 55:45 (A:B, *v*/v) at 5.0 min and was kept constant for 1 min. Afterwards, the mobile phase composition changed to 45:55 (A:B, v/v) at 6.0 min and then to 20:80 (A:B, v/v) at 7.0 min. At 8.0 min the composition of the mobile phase returned to its initial value which was maintained until 12.0 min for equilibration. The analytes were monitored at 245 nm [26]. The retention times of the pesticides were: monolinuron: 3.0 min, isoproturon: 3.7 min, linuron: 5.1 min, and chlorbromuron: 5.3 min. A representative chromatogram of the analytes is presented in Fig. 1. The gradient program may have to be adjusted if other instrumentation is used that has different dwell volumes or a different mixing technology, potentially resulting in small deviations in the retention time of the analytes.

2.4. FPSE procedure

The FPSE membranes were initially activated by immersion in 1 mL of MeOH:ACN 50:50 *v*/v for 5 min, following by immersion in 2 mL deionized water for 5 min. An aliquot of 10 mL of juice sample was transferred to a 40 mL glass vial containing a Teflon magnet. The adsorption of PUHs was conducted by immersing the FPSE membrane in the sample for 30 min under constant stirring at 500 rpm. Afterwards, the FPSE membrane was removed with tweezers and rinsed with deionized water to remove sample residues. The elution of the absorbed pesticides was conducted by immersing the FPSE membrane in 300 μL of ACN for 5 min while vortex mixing. Finally, the FPSE membrane was removed, and the eluate was analyzed by HPLC-DAD. The proposed FPSE procedure is depicted in Fig. 2. The FPSE membrane was washed by immersing in 1 mL of MeOH:ACN 50:50 *v*/v, air-dried and stored until further use.

3. Results and discussion

3.1. Optimization of FPSE conditions

The main factors that influence the efficiency of the FPSE membrane on the extraction of PUHs from fruit juice samples were examined. For this reason, the following variables were studied: type of sol-gel sorbent, size of the membrane, adsorption time, volume of the sample, stirring rate, salt addition, elution solvent, elution time and volume of the eluent. Each parameter was independently investigated while all the other factors remained stable. The initial conditions were: 2.0 cm × 2.0 cm^2 size of the membrane, 20 min adsorption time, 10 mL sample volume, 750 rpm stirring rate, 0 % *w*/*v* NaCl, 1000 μL of MeOH elution solvent, and 5 min elution time. The results were expressed as normalized extraction recoveries (ER%), calculated by the comparison of each value with the optimum one.

3.1.1. Selection of FPSE sorbent type and size

The selection of the optimum FPSE sorbent was conducted by examining different sol-gel coated FPSE membranes with different characteristics. For this reason, CW 20 M, poly(tetrahydrofuran)

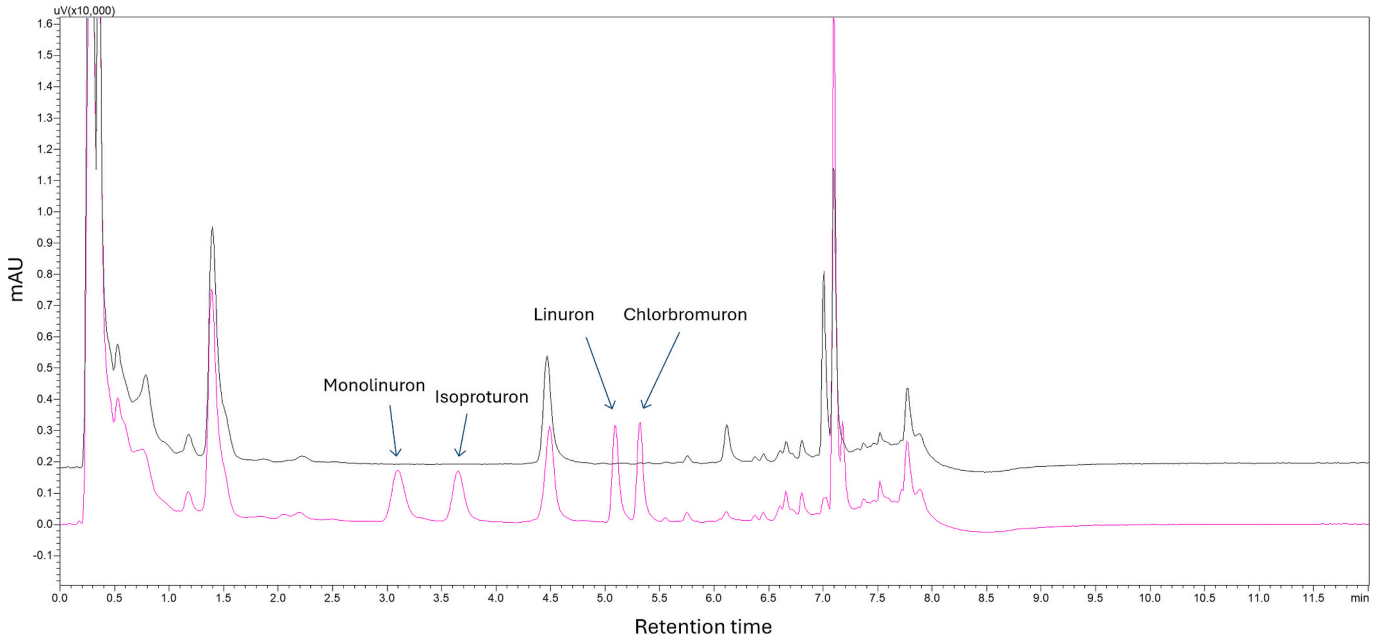


Fig. 1. Representative chromatograms of the analysis of blank and spiked apple juice sample (50 ng mL^{-1}) through the proposed FPSE-HPLC-DAD method.

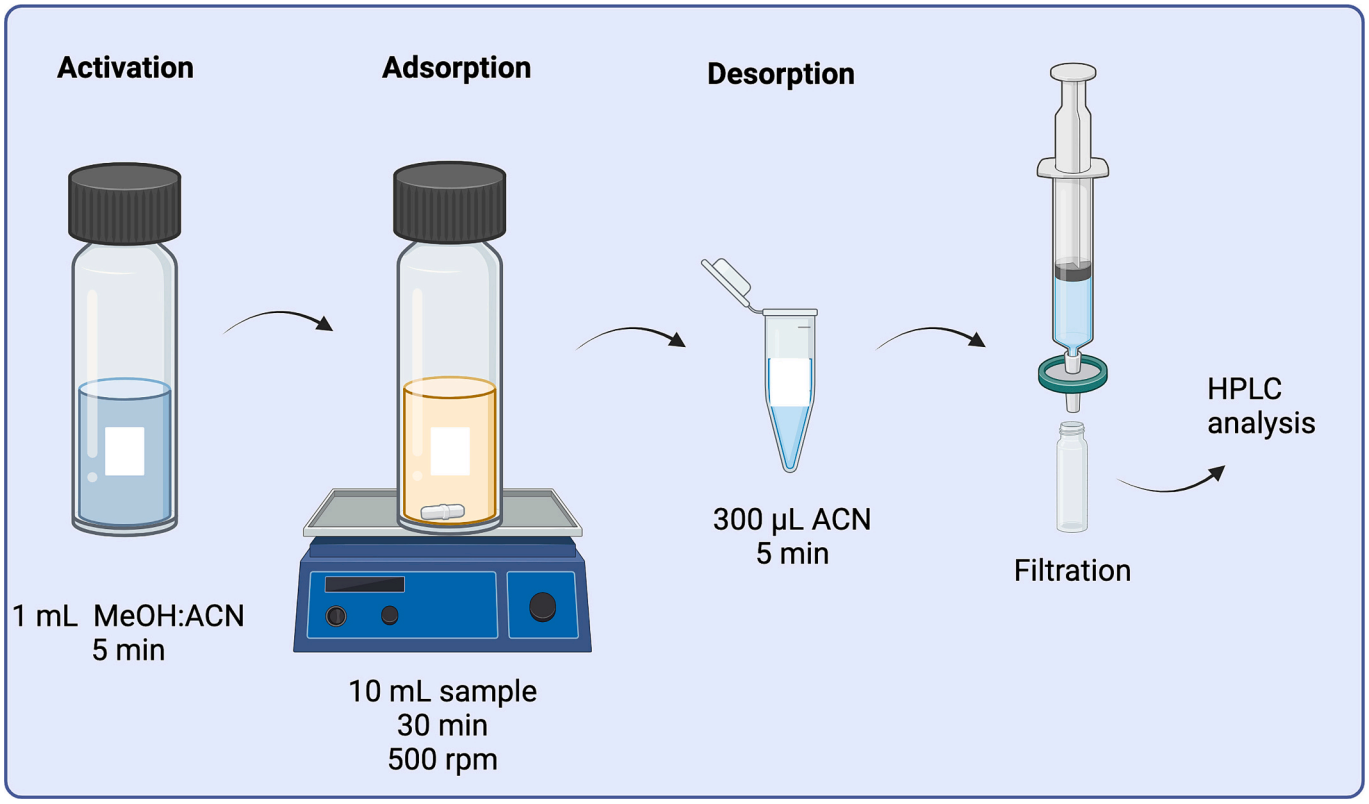


Fig. 2. Schematic depiction of the FPSE steps.

(PTHF), and octadecyl (C_{18}) membranes were used with polar, medium polar, and non-polar characteristics respectively. Regarding the target analytes, the presence of polar functional groups provides polar to medium polar characteristics. The experimental results are shown in Fig. 3. As expected, the non-polar C_{18} sorbent had limited interactions with the polar analytes, while the best results were obtained using the polar CW 20 M sorbent. This can be explained by the intermolecular forces developed between the analytes and the functional groups of the CW 20 M, mainly hydrogen bonding forces and dipole-dipole interactions. Hence, the sol–gel coated CW 20 M cellulose FPSE membrane was selected for further investigation.

The size of the membranes is a significant factor as it determines the amount of the sorbent that is available to interact with the analytes. For this reason, four different dimensions were investigated as follows: 1.0 × 1.0, 1.5 × 1.5, 2.0 × 2.0, and 2.5 × 2.5 cm^2. For this set of experiments, 300 µL of elution solvent was used to see if these dimensions are

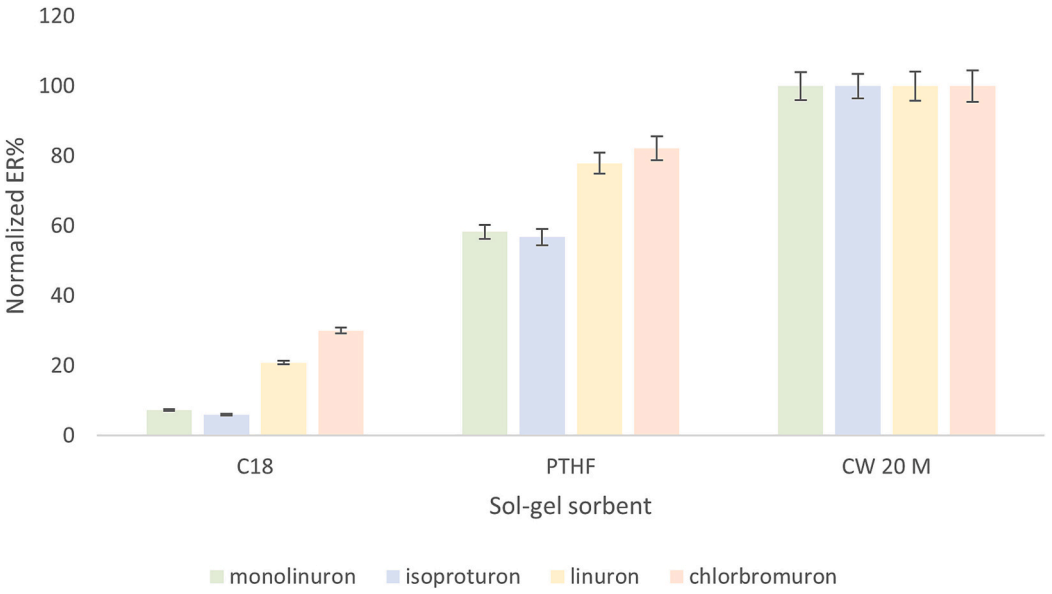


Fig. 3. Evaluation of the sol-gel sorbent membrane for the extraction of the selected pesticides. Concentration of the selected analytes: 100.0 ng mL^{-1}.

also compatible with low amounts of eluent. The results were expressed as normalized extraction recoveries (ER%). The experimental results are shown in Fig. S1. The extraction recoveries were significantly increased by increasing the dimensions in the range from 1.0 × 1.0 to 2.0 × 2.0 cm^2. However, a decrease in the extraction recoveries was observed by further increasing the dimensions of the membrane to the value 2.5 × 2.5 cm^2. When using a 2.5 × 2.5 cm^2 sized piece of FPSE, it turned out that elution was not adequately possible using the 2 mL Eppendorf vials and 300 μL elution solvent. Therefore, extraction recoveries were inadequate. Therefore, the dimension of 2.0 × 2.0 cm^2 sol-gel coated CW 20 M cellulose FPSE membrane was finally selected for further investigation.

3.1.2. *Optimization of adsorption conditions*

The adsorption time is an important parameter, as it affects the time of the analytes-sorbent interaction. FPSE is an equilibrium-based extraction process and thus the adsorption time is referred to equilibration time between the analytes and the sorbent. The adsorption time was investigated in the range 10–50 min. As shown in Fig. S2, the target analytes reached the equilibrium at 30 min. No increase was observed

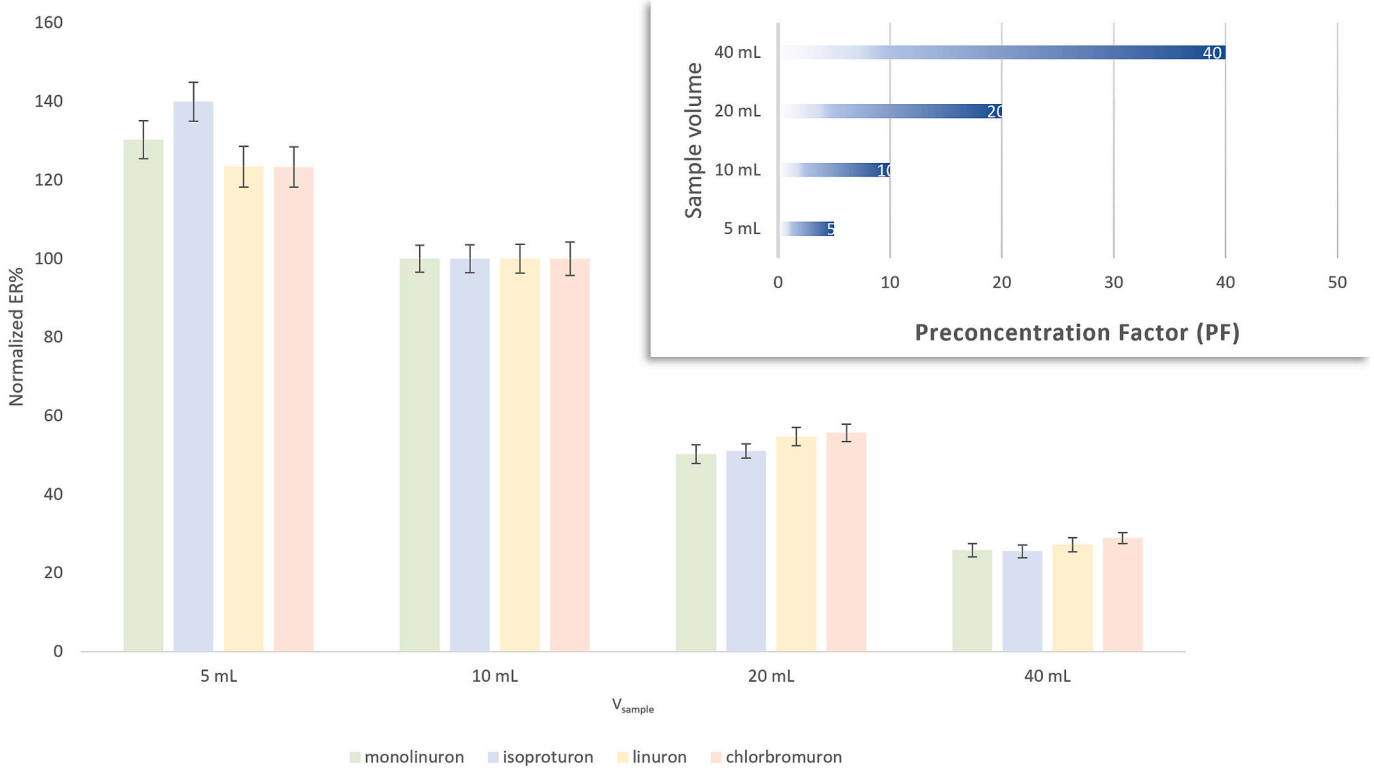


Fig. 4. Study of the sample volume for the extraction of PUHS. The inset shows the different preconentration factors (PFs) in dependence of the volume of sample processed, calculated for a final elution volume of 1 mL. Concentration of the selected analytes: 100.0 ng mL^{-1}.

with further increase of the time. Thus, 30 min was selected as the optimum adsorption time.

The volume of the sample is another important parameter because it affects the preconcentration factor and thus the sensitivity of the extraction method. The sample volume was investigated in the range 5–40 mL and the experimental results are shown in Fig. 4. As can be observed, the extraction recoveries were increased by decreasing the sample volume. However, the preconcentration factor is increased by increasing the sample volume as higher quantity of analyte is available to interact with the sorbent. Taking into account these competitive phenomena, the sample volume of 10 mL was selected as a compromise for further study.

The stirring rate was also investigated as a parameter that affects the mass transfer of the analytes to the sorbent and the adsorption. The stirring rate was studied at 0, 500, 750, and 1500 rpm. As expected, a limited amount of the analytes is adsorbed at 0 rpm. The extraction recoveries were increased at 500 rpm and were found to be stable for further increases until 1500 rpm. Hence, 500 rpm was selected as the optimum stirring rate.

The salt addition was the last parameter of the adsorption step that was examined. The ionic strength is a significant parameter that affects the adsorption through several ways. The presence of ions in the matrix can either enhance the adsorption process through the salting-out effect or decrease the analytes-sorbent interactions, because the density of the matrix is increased, influencing the mass transfer. The salt addition was investigated in the range 0–30 % *w/v* NaCl. As shown in Fig. S3, an increase of NaCl concentration caused a reduction in the extraction efficiency, indicating limited mass transfer. Consequently, no salt addition was selected for the proposed method.

3.1.3. Optimization of the desorption conditions

In order to achieve sufficient desorption of the selected pesticides from the sol-gel coated CW 20 M cellulose FPSE membrane the elution solvent was investigated. Four different solvents were examined, namely methanol (MeOH), ethanol (EtOH), acetonitrile (ACN), and acetone (ACE). The experimental results are shown in Fig. S4. As can be observed, ACN resulted in the highest extraction recoveries and thus it was selected for further study.

Another parameter that affects the preconcentration factor and the sensitivity of the method is the volume of the elution solvent. In order to achieve high preconcentration, the elution solvent has to be minimal but sufficient to successfully desorb the analytes. The elution solvent volume was investigated in the range 300–1500 μL. As shown in Fig. S5, the volume of 300 μL led to the highest extraction recoveries. In addition, this volume increased the preconcentration factor and is the most environmentally friendly selection. Therefore, 300 μL was selected as the optimum elution solvent.

Finally, the elution time was investigated. The elution time has to be sufficient to achieve effective elution of the adsorbed analytes and, at the same time, to be the minimum one in order to increase the high throughput. The elution time was examined in the range from 2 to 10 min. As shown in Fig. S6, the value of 2 min was not adequate to elute the analytes. Since the values of 5 to 10 min had comparable results, the minimum time span of 5 min was selected.

3.2. Analytical performance of the FPSE-HPLC-DAD method

The developed FPSE-HPLC-DAD method was validated in terms of linearity, limit of detection (LOD) and quantification (LOQ), trueness, and precision. Table 1 shows the analytical figures of merit of the proposed method. The linearity was assessed using a sample of organic apple juice after spiking with the target analytes at concentration levels in the range of 2.0–200.0 ng mL^{-1}. Least square linear regression analysis was used to calculate the slopes, intercepts, and coefficients of determination (R^2) of the analytes. As shown in Table 1, the r^2 values were found to be higher than 0.998 in all cases, indicating excellent linearity of the targeted analytes.

The LOQ values of PUHs were defined as the lowest points of the calibration curves and were calculated as 2.0 ng mL^{-1}. The LOD values were calculated by dividing the LOQ values by 3.3. Therefore, the LODs were found to be 0.6 ng mL^{-1}.

The enhancement factors (EF) were calculated after the comparison of the slopes of the calibration curve for the sample solutions after the FPSE procedure and the slope of the calibration curve of standard aqueous solutions that corresponded to the same concentration levels. The (theoretical) preconcentration factor (PF) was assessed by dividing the initial volume of the sample (10 mL) and the final volume of the sample (300 μL) and was calculated to be 33. The extraction recoveries were calculated by comparing the EF with the PF of each analyte. The ER % of the PUHs ranged between 58 and 60 %, while their enhancement factors ranged between 19 and 20.

The precision and trueness of the proposed method were examined using samples of organic apple juice spiked at two concentration levels (low = 5.0 ng mL^{-1} and high = 50.0 ng mL^{-1}). Table S1 summarizes the results of the trueness and precision studies. The intra-day precision was calculated by successive injections ($n = 4$) and expressed as the relative standard deviation (RSD%). The inter-day precision was evaluated by the analysis of the spiked samples in three different days. The RSD values ranged from 3.5 to 7.3 % for intra-day precision and between 6.4 and 12.8 % for inter-day precision. The trueness of the method was expressed as the relative recovery (RR%) and calculated by the comparison of the found to the added concentration of the spiked samples. The RR% was satisfactory and ranged from 84.5 to 109.8 % for the intra-day accuracy and between 94.5 % and 113.5 % for the inter-day accuracy.

3.3. FPSE membrane reusability and carry over

Initially, potential carry-over was checked by employing the same membrane for the extraction of a mixture of PUHs at 200 μg L^{-1}, followed by blank analysis. Since no PUHs were detected during the analysis of the blank sample, the FPSE membrane was tested for its reusability by repeating the extraction of PUHs from a spiked apple juice sample at 50.0 ng mL^{-1}. For this purpose, twenty consecutive extractions were performed using the same FPSE medium and a deviation of up to 15 % was considered acceptable for the loss of extraction efficiency in comparison with its initial performance. As shown in Fig. S7, after the twenty repeated extractions the RR% were found to be in the range 85–89 %. Consequently, the sol-gel CW 20 M coated FPSE membrane can be applied for the extraction of the selected pesticides from apple juice samples for twenty times, being in accordance with the requirements of GSP for using reusable materials.

Table 1
Analytical figures of merit of the proposed FPSE-HPLC-DAD method.

Analyte	Regression Analysis	R^2	Linear range (ng mL^{-1})	LOD[1] (ng mL^{-1})	LOQ[2] (ng mL^{-1})	ER%[3]	EF[4]
Monuron	y = 265x – 480	0.9995	2.0–200	0.6	2.0	58	19
Isoproturon	y = 211x + 357	0.9988	2.0–200	0.6	2.0	58	19
Linuron	y = 270x – 257	0.9996	2.0–200	0.6	2.0	60	20
Chlorbromuron	y = 266x – 298	0.9997	2.0–200	0.6	2.0	60	20

3.4. Comparison with other studies

The developed method was compared with other studies that use extraction or microextraction techniques and HPLC for the determination of PUHs in similar matrices. The results are shown in Table 2. Among the methods listed, the proposed method has satisfactory sensitivity with similar LOD values to Chen et al. [27] and Legesse et al. [28], and better than the methods [6,29,30]. Regarding the trueness and precision of the proposed method, the results are comparable with the other methods. The enhancement factors of Chen et al. [27] and Legesse et al. [28] are better than the developed method, while the other methods do not mention their enhancement factors. It should be noted that the present FPSE method was used to extract a larger number of analytes compared to the majority of the other studies [6,27,29,30]. The results demonstrated that the developed method is efficient for the determination of PUHs in fruit juice samples.

3.5. Real samples analysis

The proposed FPSE-HPLC-DAD method was applied for the determination of the targeted PUHs in apple juice samples. Table S2 summarizes the results of real sample analysis. The examined PUHs were not detected in any of the samples. Spiked samples were also prepared and analyzed at the concentration level of 50.0 μg L^{-1}. The RR% for each analyte was calculated. As it can be observed, the RR% values for all analytes were found to be in the range of 88.1–114.0 %, indicating good trueness.

3.6. Examination of the green character and practicality of the developed procedure

The applicability of the proposed method in terms of practicality was examined using BAGI [21]. This metric tool evaluates the instrumental analysis and the sample preparation technique that is required, and it produces an asteroid pictogram with its parts corresponding to an individual criterion. These parts can have a dark blue, blue, light blue, and white color that reveal good, medium, low, and no compliance with practicality requirements. Moreover, a numerical score between 25.0 and 100.0 is calculated, which should be higher than 60.0 to demonstrate good practicality. The FPSE-HPLC-DAD method exhibited a BAGI score of 72.5 showing good practicality. The BAGI pictogram is depicted in Fig. 5a. This can be attributed to the good method sensitivity that eliminated any need for additional preconcentration, the possibility of parallel sample handling, and the commonly available instrumentation.

At a final step, the green character of the developed method was assessed. Currently, multiple green profile tools have been developed and are used for the evaluation of the environmental impact of the method [31]. In this work, ComplexGAPI index, ComplexMoGAPI and CACI were employed to study method's greenness. ComplexGAPI examines the environmental impact of the analytical method by evaluating different parameters (i.e., instrumentation, sample collection/pretreatment, method type, quantification, reagents/solvents, and sample preparation) [22]. At the same time, it can examine the compliance of the preparation of the FPSE membranes with greenness requirements. A pictogram is produced with a color coding of red, yellow, green for low, medium, and high compliance. Fig. 5b shows the obtained pictogram for the FPSE-HPLC-DAD method. As can be observed, the developed method complies with some of the requirements due to the elimination of the use of chemicals, the reduction of waste generation and energy requirements. ComplexMoGAPI is a recently introduced modification of ComplexGAPI that combines its visual appeal with a precise total score, towards a more quantitative greenness assessment [23]. A score of 78 was produced using ComplexMoGAPI as shown in Fig. 5c, revealing the good environmental compliance of the proposed method. Finally, CACI evaluation was also performed [24]. This index examines the green merits of sample size, sample preparation, feasibility, application, portability, automation, and sensitivity of an analytical method. CACI supports approaches that simplify the analytical process, as it is inspired by the simplicity and reliability of click chemistry. A numerical score is also generated with this metric tool for a quantitative method evaluation. As shown in Fig. 5d, a score of 62 was obtained.

4. Conclusions

In this work, polar FPSE membranes consisting of a sol–gel CW 20 M sorbent and cellulose substrate were used for the extraction of PUHs from fruit juice samples prior to their quantitation by HPLC-DAD. The developed protocol exhibited handling simplicity, practicality, greenness, and rapidity. The FPSE membranes were reusable for up to 20 times, and multiple samples could be treated simultaneously. The analytical method exhibited good performance characteristics in terms of sensitivity, linearity, trueness, and precision. Moreover, good applicability was demonstrated for apple juice analysis. In case of the analysis of other fruit juices or similar samples (e.g., tea, beverages, herbal

Table 2
Comparison of the proposed FPSE-HPLC-DAD method with other HPLC methods for the determination of pesticides in fruits or fruit juice samples.

PUHs	Sample preparation	Analytical technique	Sample	RR %	RSD %	EF	LOD (ng mL^{-1} or ng g^{-1})	Ref.
Linuron	Effervescence-assisted dispersive liquid-liquid microextraction	HPLC-DAD	Fruit juices (8 mL)	72.18–109.15	< 3.23 (intra) < 4.6 (inter)	153.85	0.16	[27]
Monuron, chlortoluron, isoproturon, and monolinuron	Dispersive liquid-liquid microextraction using supramolecular solvent	HPLC-DAD	Fruit juices (10 mL)	81.21–113.12	< 4.99 (intra) < 5.98 (inter)	37.1–72.8	0.13–0.19	[28]
Thidiazuron and forchlorfenuron	Ultrasound-assisted dispersive-filter extraction	HPLC-UV	Vegetables and fruits (25 g)	86.8–96.8	< 5.4 (intra) < 7.1 (inter)	N.A.	1.0–1.25	[29]
Monuron, chlortoluron, and monolinuron	In-syringe temperature-controlled liquid–liquid microextraction	HPLC-DAD	Vegetable protein drinks (8 mL)	97.29–103.56	< 4.90 (intra) < 5.16 (inter)	N.A.	0.25–2.59	[30]
Monolinuron, diuron and diethofencarb	Solid phase microextraction	LC-MS	Fruit juices (0.5 mL)	95–105	< 15 (intra and inter)	N.A.	5–10	[6]
Monolinuron, linuron, isoproturon, and chlorbromuron	FPSE	HPLC-DAD	Fruit juices (10 mL)	84.5–109.8	< 7 (intra) < 13 (inter)	19–20	0.6	This study

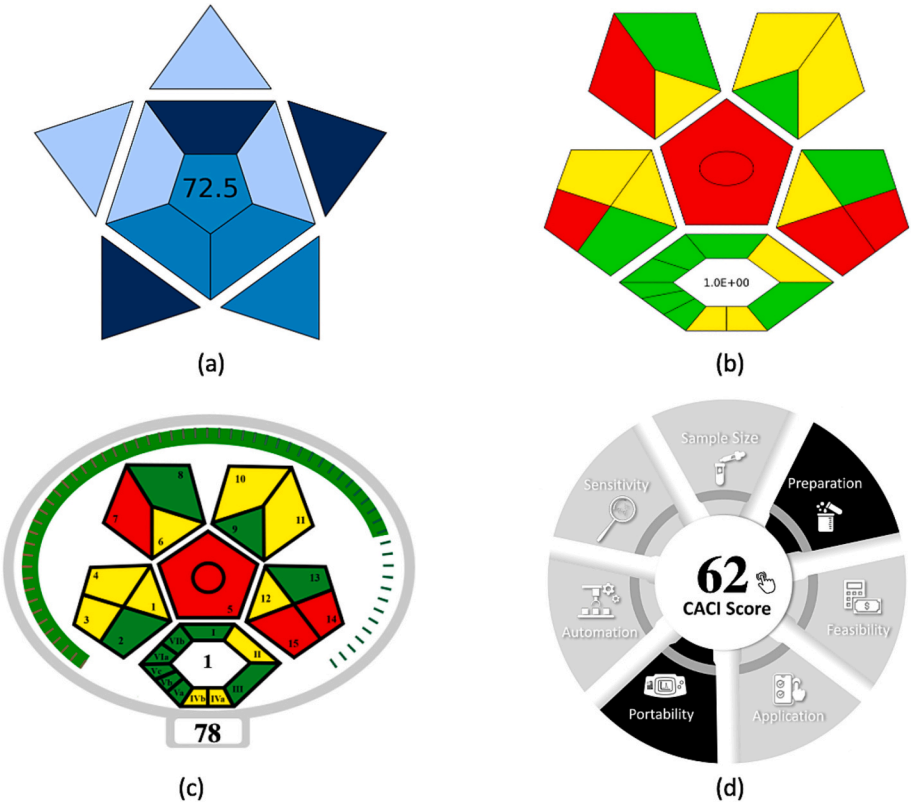


Fig. 5. BAGI *(a)*, ComplexGAPI *(b),* ComplexMoGAPI (*c*), and CACI (*d*) pictograms of the proposed method.

infusions etc.), the proposed method can be modified appropriately (e.g., by using matrix-matched calibration curves) and its applications can be significantly expanded. All things considered, FPSE can be used as a simple and robust tool in apple juice analysis for PUHs pesticide residue monitoring.

CRediT authorship contribution statement

Apostolia Tsiasioti: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Natalia Manousi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Abuzar Kabir:** Writing – review & editing, Resources, Methodology, Conceptualization. **Erwin Rosenberg:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

All authors declare that they have no competing financial interests or personal reationships that could have appeared to influence the reported work in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.microc.2025.114867.

Data availability

Data will be made available on request.

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